Econometrics II – Assignment 1: Forecasting Economic Growth and Growth-at-Risk

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1 Introduction

This paper proposes a univariate time series model for U.S. quarterly GDP growth and uses it to produce one-step-ahead point forecasts and Growth-at-Risk (GaR) estimates at the 10% risk level. Understanding the conditional distribution of future output growth is of direct relevance for macroprudential policy and central banking, where downside risk to GDP is a key input to stress-testing frameworks (Adrian, Boyarchenko, and Giannone, 2019).

We estimate autoregressive models of order one through four, as well as an ARMA(1,1), on U.S. data for the period 1990Q1–2009Q4, and evaluate out-of-sample performance over 2010Q1–2024Q4 (60 observations). Model selection based on the Bayesian Information Criterion (BIC) and Ljung-Box diagnostic tests points to an AR(2) as the preferred specification. The model is then used to produce fixed-parameter one-step-ahead forecasts and to derive Gaussian conditional GaR estimates.

Our main findings are: (*i*) the AR(2) forecasts are not statistically different from a naive historical-mean benchmark (Diebold-Mariano p-value = 0.30); and (*ii*) neither the AR(2) GaR nor the unconditional GaR benchmark can be rejected for correct conditional coverage at the 10% level. These results are consistent with recent evidence that simple univariate models remain competitive with more sophisticated approaches (Adrian et al., 2025).

2 Description of the Data

The data consist of quarterly observations from 1990Q1 to 2024Q4. The variable of interest is $y_t = \log(\text{GDPC1}_t) - \log(\text{GDPC1}_{t-4})$, the log year-on-year growth rate of U.S. Real GDP (GDPC1), retrieved from the FRED database. Using a year-on-year transformation removes seasonal effects without requiring seasonal adjustment.

Figure 1 plots the full sample. The series fluctuates around a positive mean with no apparent deterministic trend, consistent with stationarity. Two episodes stand out: growth contracted sharply during the Global Financial Crisis (2008–2009) and again during the COVID-19 pandemic (2020Q2), which produced an extreme outlier of roughly -9% followed by an equally sharp V-shaped recovery. Outside these episodes, the series is relatively smooth and mean-reverting, motivating the use of a low-order autoregressive model.

Figure 2 shows the sample autocorrelation function (ACF) and partial autocorrelation function (PACF) for the training period 1990Q1–2009Q4. The ACF decays gradually over several lags, indicating persistence, while the PACF drops sharply after lag 2 with remaining partial autocorrelations close to zero. This pattern is the classic signature of an AR(2) process, and guides our subsequent model selection.

Figure 1: U.S. quarterly GDP growth y_t, 1990Q1–2024Q4. Shaded region indicates the out-of-sample evaluation period (2010Q1–2024Q4).

3 Econometric Theory

3.1 The ARMA Model

We consider the ARMA(p,q) model

$$y_t = \delta + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}, \quad \varepsilon_t \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2), \tag{1}$$

where y_t is the log year-on-year GDP growth rate and δ is the intercept. A pure AR(p) corresponds to $q = 0$. Stationarity requires that all roots of the AR polynomial $1 - \phi_1 L - \cdots - \phi_p L^p = 0$ lie outside the unit circle, under which the unconditional mean equals $\mu = \delta/(1 - \phi_1 - \cdots - \phi_p)$. In practice, statistical software such as statsmodels parameterises the model directly in terms of μ, so the reported constant is the unconditional mean, not the intercept δ.

3.2 Estimation

For a pure AR(p) model, the parameters $\boldsymbol{\phi} = (\delta, \phi_1, \ldots, \phi_p)'$ are estimated by OLS, minimising $\sum_{t=p+1}^{T} \varepsilon_t^2(\boldsymbol{\phi})$. Under stationarity and ergodicity, and given the assumption $E[\varepsilon_t \mid \mathcal{I}_{t-1}] = 0$, the OLS estimator is consistent and asymptotically normal. Specifically,

$$\sqrt{T}\,(\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}) \xrightarrow{d} N\big(\mathbf{0},\, \sigma^2 \Gamma_p^{-1}\big)\,,$$

where Γ_p is the $(p+1) \times (p+1)$ matrix of autocovariances. For the ARMA(p,q) model with $q \geq 1$, OLS is infeasible because the MA errors are unobserved; estimation proceeds via Gaussian maximum likelihood (ML), maximising

$$\ell(\boldsymbol{\phi}, \boldsymbol{\theta}, \sigma^2) = -\tfrac{T}{2}\log(2\pi\sigma^2) - \tfrac{1}{2\sigma^2}\sum_{t=1}^{T} \varepsilon_t^2,$$

where ε_t is obtained recursively from the observed data. Note that OLS is numerically equivalent to ML for the AR(p) case.

Model selection proceeds by first comparing information criteria — the Akaike criterion $\mathrm{AIC} = -2\ell + 2k$ and the Bayesian criterion $\mathrm{BIC} = -2\ell + k\log T$, where k is the number of estimated parameters — and then verifying that the preferred model passes residual diagnostic tests. BIC is preferred over AIC because its heavier penalty on k yields more parsimonious models in finite samples.

3.3 Forecasting

The one-step-ahead forecast from model (1) is the conditional mean

$$\hat{y}_{t+1} = E[y_{t+1} \mid \mathcal{I}_t] = \delta + \phi_1 y_t + \cdots + \phi_p y_{t-p+1} + \theta_1 \varepsilon_t + \cdots + \theta_q \varepsilon_{t-q+1}.$$

Parameters are fixed at their training-sample estimates. Residuals $\hat{\varepsilon}_t$ required for the MA terms are updated recursively via $\hat{\varepsilon}_t = y_t - \hat{y}_t$ for $t > T$. For a pure AR model the MA component is absent and the forecast reduces to $\hat{y}_{t+1} = \delta + \phi_1 y_t + \cdots + \phi_p y_{t-p+1}$.

Forecast accuracy is assessed against the naive benchmark $\bar{y} = T^{-1}\sum_{t=1}^{T} y_t$ using root mean squared error (RMSE). Statistical significance of any difference is evaluated by the Diebold-Mariano test: define $d_t = e_{\text{model},t}^2 - e_{\text{naive},t}^2$ where $e_{j,t} = y_t - \hat{y}_{j,t}$; then under $H_0\colon E[d_t] = 0$ the test statistic

$$t_{\mathrm{DM}} = \frac{\bar{d}}{s_d/\sqrt{T}} \xrightarrow{d} N(0,1),$$

where $\bar{d}$ and s_d are the sample mean and standard deviation of (d_t).

3.4 Growth-at-Risk

The one-period Growth-at-Risk at risk level $\alpha \in (0,1)$, denoted GaR_t^α, is the α-quantile of the conditional distribution of y_t given $\mathcal{I}_{t-1}$:

$$P(y_t \leq \mathrm{GaR}_t^\alpha \mid \mathcal{I}_{t-1}) = \alpha.$$

Under the Gaussian assumption $\varepsilon_t \sim N(0, \sigma^2)$, this simplifies to

$$\mathrm{GaR}_t^\alpha = \hat{y}_t + \sigma\Phi^{-1}(\alpha), \tag{2}$$

where $\Phi^{-1}(\alpha)$ is the α-quantile of the standard normal. For $\alpha = 0.1$, $\Phi^{-1}(0.1) \approx -1.282$.

Evaluation uses the hit indicator $\mathrm{hit}_t = \mathbf{1}\{y_t < \mathrm{GaR}_t^\alpha\}$ and the violation rate $\hat{p} = T^{-1}\sum_t \mathrm{hit}_t$. Under correct specification (hit_t) is i.i.d. Bernoulli(α) and we test $H_0\colon E[\mathrm{hit}_t] = \alpha$ using the t-statistic $t = (\hat{p} - \alpha)/\sqrt{\alpha(1-\alpha)/T}$, which is asymptotically $N(0,1)$. Additional discriminating power is provided by the tick loss

$$\mathcal{T}_\alpha(y_t, \mathrm{GaR}_t^\alpha) = (\alpha - \mathrm{hit}_t)(y_t - \mathrm{GaR}_t^\alpha),$$

which weighs under-predictions more heavily than over-predictions.

4 Empirical Analysis

4.1 Model Selection

Figure 2 motivated a focus on AR models: the PACF is statistically significant at lags 1 and 2 and negligible thereafter, suggesting that an AR(2) is a natural starting point. We estimate AR(1)–AR(4) and ARMA(1,1) and compare them in Table 1.

The AR(1) is immediately rejected: the Ljung-Box p-value at lag 8 equals 0.00, indicating that significant residual autocorrelation remains. All higher-order models pass the Ljung-Box test. Among these, the AR(2) achieves the best BIC (-542.33), reflecting a good balance between fit and parsimony. Moving to AR(3) raises the AIC marginally (-552.15 vs. -551.86) but worsens the BIC (-540.24 vs. -542.33) because the additional parameter is not justified relative to its penalty of $\log T \approx 4.4$. AR(4) and ARMA(1,1) are also dominated by AR(2) on BIC. We therefore select **AR(2)** as the preferred specification.

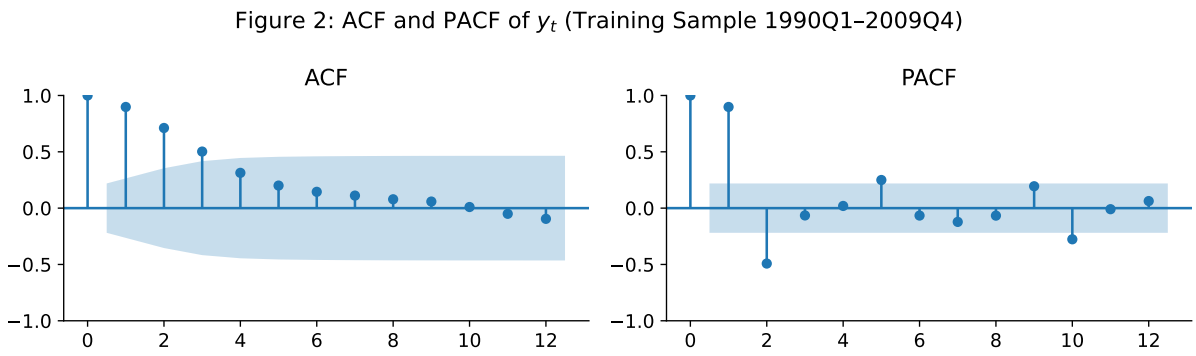

Figure 2: Sample ACF and PACF of y_t, training sample 1990Q1–2009Q4. Dashed lines indicate 95% confidence bands.

Table 1: Model comparison: AIC, BIC, $\hat{\sigma}$, and Ljung-Box p-value (training sample 1990Q1–2009Q4).

Model	AIC	BIC	$\hat{\sigma}$	LB p-val (lag 8)
AR(1)	-535.78	-528.64	0.01	0.00
AR(2)	-551.86	-542.33	0.01	0.14
AR(3)	-552.15	-540.24	0.01	0.24
AR(4)	-550.23	-535.94	0.01	0.28
ARMA(1,1)	-546.13	-536.60	0.01	0.12

4.2 Misspecification Testing

Before interpreting the estimated coefficients we assess whether the AR(2) residuals satisfy the assumptions required for valid inference. Three tests are applied to the training-sample residuals.

The Ljung-Box test for residual serial correlation yields $Q(4) = 8.67$ ($p = 0.070$) and $Q(8) = 12.25$ ($p = 0.140$). Both p-values exceed the conventional 5% significance level, so we do not reject the null hypothesis of no residual autocorrelation up to the tested lag. The marginally borderline lag-4 result is consistent with a mild GFC-era clustering of large residuals rather than systematic misspecification.

The Jarque-Bera test for normality yields JB $= 8.63$ ($p = 0.013$), rejecting the null of normally distributed residuals at the 5% level. Inspection of the residuals reveals a mildly fat-tailed distribution (excess kurtosis ≈ 1.53), attributable in part to the relatively large residuals around the GFC. The non-normality does not invalidate OLS consistency or standard errors (by the CLT these are asymptotically valid under stationarity and ergodicity), but it implies that the Gaussian GaR formula in equation (2) may not precisely calibrate the lower tail.

The ARCH-LM test at four lags yields a statistic of 7.98 ($p = 0.092$), failing to reject the null of no conditional heteroskedasticity at the 5% level. There is therefore no strong evidence of time-varying volatility in the quarterly training data.

4.3 Preferred Model: Estimates and Interpretation

Table 2 reports the ML estimates for the AR(2) model over 1990Q1–2009Q4. The estimated unconditional mean is $\hat{\mu} = 0.0262$ (standard error 0.006), corresponding to an average annual GDP growth of approximately 2.6%, which is in line with the U.S. long-run average. Both AR coefficients are highly significant: $\hat{\phi}_1 = 1.359$ ($t = 11.0$) and $\hat{\phi}_2 = -0.512$ ($t = -4.1$). The sum $\hat{\phi}_1 + \hat{\phi}_2 = 0.847$ is below unity, confirming stationarity. The pair of AR roots are complex conjugates with modulus $\sqrt{|\hat{\phi}_2|} \approx 0.72 < 1$, consistent with stationary damped oscillations around the mean. The estimated residual standard deviation is $\hat{\sigma} = 0.010$, which enters the GaR formula as the scale of the conditional distribution.

Table 2: Estimation results for the AR(2) model (training sample 1990Q1–2009Q4).

	Coefficient	Std. Error	t-stat	p-value
const	0.0262	0.0060	4.3455	0.0000
ar.L1	1.3587	0.1235	11.0055	0.0000
ar.L2	-0.5118	0.1241	-4.1237	0.0000
sigma2	0.0001	0.0000	6.5740	0.0000

4.4 Forecast Evaluation

Figure 3 plots the realised GDP growth alongside the AR(2) and naive forecasts over 2010Q1–2024Q4. Both forecasts track the broad cyclical movements reasonably well, but neither anticipates the extreme COVID-19 trough in 2020Q2. Table 3 summarises the evaluation metrics.

The AR(2) RMSE is 0.0256 compared to 0.0206 for the naive benchmark. The naive forecast is actually lower, which at first glance suggests an advantage for the simpler model. However, the Diebold-Mariano test statistic $t_{\mathrm{DM}} = 1.045$ ($p = 0.296$) fails to reject the null hypothesis of equal mean squared error. We therefore cannot conclude that the AR(2) significantly underperforms the naive benchmark; the difference is driven almost entirely by the model's inability to predict the COVID-19 shock, which lies outside the dynamics captured by any linear AR model. This result aligns with recent findings that simple linear time-series models are not systematically dominated by more complex approaches.

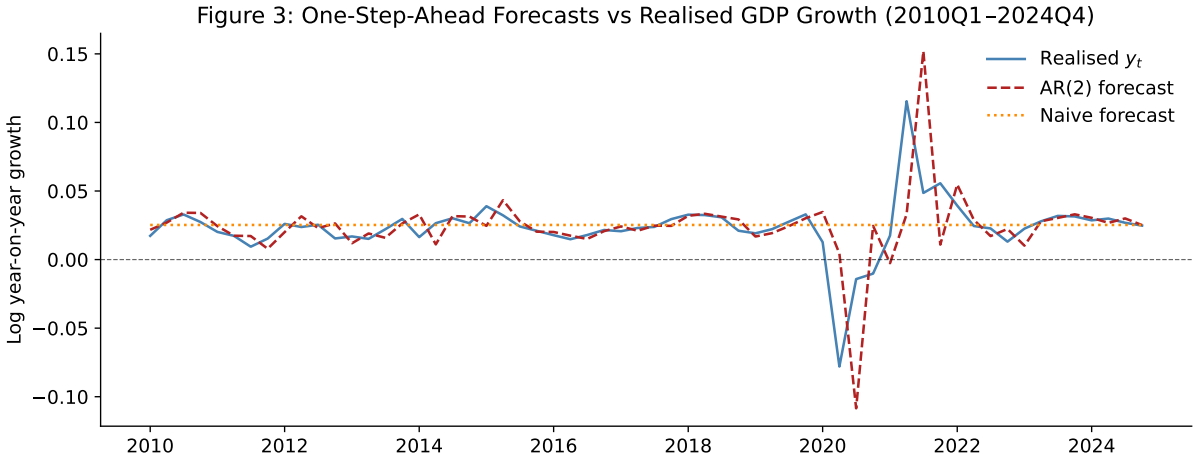

Figure 3: Out-of-sample forecasts vs realised GDP growth, 2010Q1–2024Q4.

Table 3: Forecast evaluation: RMSE and Diebold-Mariano test (2010Q1–2024Q4).

Model	RMSE	DM statistic	DM p-value
AR(2)	0.0256	1.0450	0.2960
Naive	0.0206	–	–

4.5 Growth-at-Risk Evaluation

Figure 4 plots the AR(2) GaR (10% level) alongside the unconditional GaR benchmark (GaR^{np}) and realised GDP growth over 2010Q1–2024Q4. The AR(2) GaR is time-varying, tracking the level of the forecast, while GaR^{np} is a constant at the 10th percentile of the training-sample distribution. Table 4 presents the evaluation statistics.

The AR(2) GaR has a violation rate of $\hat{p} = 0.150$, exceeding the target of 10%. The coverage t-test yields $t = 1.291$ ($p = 0.197$), so we cannot reject correct coverage at conventional significance levels. The GaR^{np} series shows the opposite pattern: $\hat{p} = 0.050$, with the same t-statistic in absolute value ($p = 0.197$), also not rejected. Both series deviate from the 10% target in opposite directions, but neither deviation is statistically significant given only 60 evaluation observations. The AR(2) excess violations are largely attributable to the two COVID-19 quarters in 2020, when growth fell well below any pre-pandemic conditional forecast. In terms of tick loss, GaR^{np} achieves a marginally lower mean loss (0.0040 vs. 0.0046), though the difference is small. Overall, the AR(2) GaR provides useful time-variation relative to the constant benchmark, but does not achieve demonstrably better calibration over this particular evaluation window.

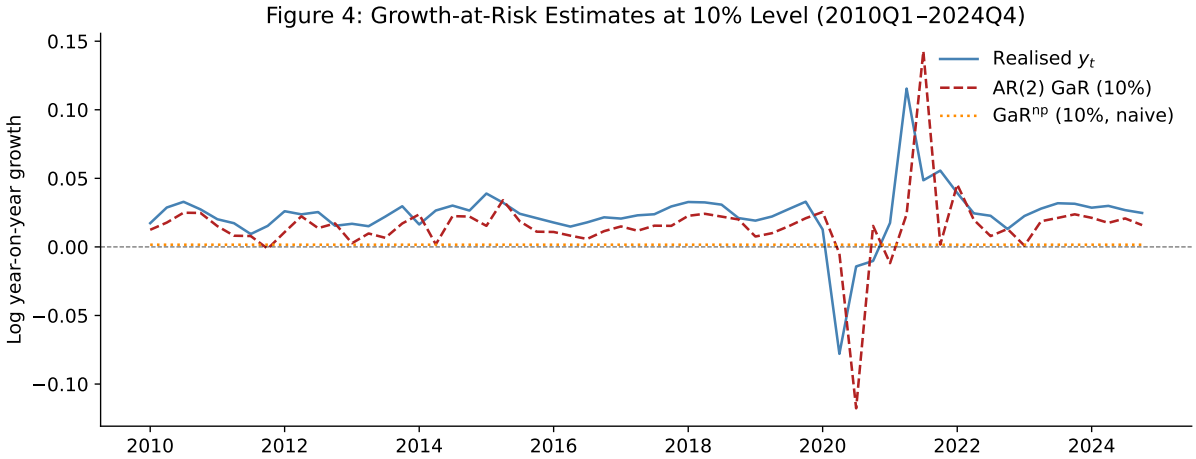

Figure 4: Growth-at-Risk estimates (10% level) vs realised GDP growth, 2010Q1–2024Q4.

Table 4: GaR evaluation at 10% level: violation rates, coverage test, and tick loss (2010Q1–2024Q4).

GaR series	Violation rate $\hat{p}$	t-stat ($H_0\colon \hat{p} = 0.10$)	p-value	Mean tick loss
AR(2) GaR	0.1500	1.2910	0.1970	0.0046
GaR^{np} (naive)	0.0500	-1.2910	0.1970	0.0040

5 Discussion

Several limitations of the approach deserve mention. First, the Jarque-Bera test rejects the normality of the AR(2) residuals. Because the GaR estimate in equation (2) is derived under normality, the tail probabilities may be mis-calibrated. A robust alternative would use Student-t or skew-normal distributed errors, which can better account for the fat-tailed, slightly asymmetric distribution of GDP growth, particularly during crisis episodes.

Second, the parameters are fixed at their training-sample estimates throughout the 2010–2024 evaluation period. No information from the evaluation sample is used to update the model, so any structural change in the dynamics of GDP growth — such as the low-volatility expansion of the 2010s or the post-COVID recovery patterns — is not captured. Recursive or rolling re-estimation would allow the model to adapt, at the cost of additional complexity.

Third, the model is purely univariate. Macro-financial variables such as credit spreads, term spreads, or financial conditions indices may contain incremental predictive content for both the conditional mean and, especially, the conditional tail of GDP growth, as documented in the GaR literature. Incorporating such predictors within a mixed-data or quantile regression framework may improve both point forecasts and GaR estimates.

Finally, the evaluation window contains the COVID-19 pandemic, an unprecedented external shock that is orthogonal to any historically estimated AR model. This single event has an outsized influence on both RMSE comparisons and GaR violation counts. A fair assessment of model performance may benefit from robustness checks that exclude or down-weight extreme outlier periods.

6 Conclusion

This paper proposed an AR(2) model for U.S. quarterly GDP growth, selected on the basis of BIC and Ljung-Box residual diagnostics from a candidate set of AR(1)–AR(4) and ARMA(1,1) models. The estimated unconditional mean of 2.6% per annum and the high AR persistence ($\hat{\phi}_1 + \hat{\phi}_2 = 0.847$) are economically plausible. The model passes standard diagnostic tests for residual autocorrelation and conditional heteroskedasticity, though the Gaussian assumption is rejected by the Jarque-Bera test.

Out-of-sample, the AR(2) forecasts are statistically indistinguishable from the naive historical-mean benchmark, with a Diebold-Mariano p-value of 0.30. For GaR, both the AR(2) and the unconditional benchmark pass the 10% coverage test; neither clearly dominates on tick loss over the 60-quarter evaluation window. The COVID-19 shock is the dominant driver of deviations in both exercises. These results are consistent with the broader empirical finding that simple linear univariate models remain competitive in macroeconomic forecasting and risk quantification.

References

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